

# Karan Singh

## Lead AI/ML Engineer

[karan963.in](mailto:karan963.in) | +91 7973742385 | [karansinghthundu@gmail.com](mailto:karansinghthundu@gmail.com) | HP, India

[leetcode.com/u/karan\\_963](https://leetcode.com/u/karan_963) | <https://github.com/karan-963>

## Summary

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Lead AI/ML Engineer with expertise in designing, training, and deploying advanced Artificial Intelligence systems, including custom Computer Vision pipelines, robust NLP architectures, and Agentic AI. Specialized in Edge AI optimization, model quantization, and end-to-end full-stack integration. Proven ability to architect complex solutions from independent research to enterprise-grade deployment.

## Experience

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### ML Engineer — *Freelancing*

October 2025 – Present

- **AI Architecture (Innova Amtech):** Led AI system design and architecture for Innova OS, focusing on scalable intelligence integration and low-latency processing.
- **Advanced AI Research:** Conducted independent research and development, resulting in the creation of two flagship machine learning pipelines: an Edge AI Computer Vision model (Hybrid99) and an advanced NLP Information Extraction engine (Resume TLM).
- **Technical Consulting (Thinkuvate):** Engineered robust application architectures and backend systems as a technical consultant, ensuring high-performance infrastructure without core AI dependencies.

### Lead AI Engineer — *Kuanta AI*

April 2025 – September 2025

- Led the development and deployment of agentic AI systems, custom training modules, and intelligent chatbot architectures.
- Integrated advanced large language models into workflows, improving complex data retrieval and automating decision-making processes.
- Designed high-performance AI solutions, collaborating across teams to transition from research into scalable enterprise applications.

### Sr. Full Stack AI Engineer — *Weblianz Solutions*

September 2022 – January 2025

Architected and maintained scalable full-stack AI applications utilizing NestJS and MongoDB, specializing in the development of intelligent chatbots and RAG (Retrieval-Augmented Generation) systems.

### Founder — *Scaptor*

August 2020 – September 2022

Founded and led development of full-stack applications using Next.js, React Native, Flutter, and cloud infrastructure, delivering scalable software solutions across web and mobile platforms.

## Skills

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**AI/ML Domain :** Computer Vision (Segmentation & Classification), NLP, Information Extraction, Edge AI, Quantization Aware Training (QAT), INT8 Quantization, RAG Architectures, Agentic AI.

**Models & Frameworks :** DistilRoBERTa-Base, SAM2 (Segment Anything Model 2), CNN Architectures, PyTorch, TensorFlow, MLX, Huggingface, Langchain

**Programming Languages :** TypeScript, Python, C++, Rust, SQL

**Cloud & Infrastructure:** AWS, GCP, Azure, Docker, Vector Databases, Cloud Architecture

## Projects

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**Resume TLM** — *Advanced NLP Information Extraction Engine*

<https://karan963.in/products/resume-tlm>

- Architected a decoupled, three-phase NLP pipeline encompassing boundary detection, semantic sequence classification, and a sophisticated rules engine for high-precision document parsing.
- Integrated and fine-tuned **DistilRoBERTa-Base** for robust spatial fusion and highly accurate semantic sequence classification.
- Combined large language model (LLM) verification with custom deterministic heuristics to achieve superior entity extraction and minimize hallucination across complex layouts.

**Hybrid99** — *Edge AI Vision System for Grain Analysis*

<https://karan963.in/products/hybrid99>

- Designed and trained a highly optimized "Detect-Split-Classify" Computer Vision pipeline for automated rice grain quality detection.
- Utilized **SAM2 (Segment Anything Model 2)** for advanced mask generation and high-quality synthetic training data preparation.
- Engineered custom sub-5MB vision models specifically optimized for edge devices, leveraging **INT8 Quantization**, Depthwise Separable Convolutions, and **Quantization Aware Training (QAT)** to maintain high accuracy at low latency.

**Super Mate** — Educational Organization Management System

<https://supermate.org>

Super Mate is a student record management desktop app built to manage all student records & other document related stuff at one place.

**CV Score Generator AI**

To generate a score out of 100 based on job description. [check demo](#)

**Knowledgebase**

Chat with your custom data or website. [check demo](#)